

The Role of Cow's Milk Allergy in Pediatric Chronic Constipation: A Randomized Clinical Trial

Objective Cow's milk allergy has different presentations in children and can cause functional bowel symptoms such as chronic constipation. The aims of this study were to investigate the role of cow's milk allergy as a cause of chronic constipation and effect of cow's milk free diet (CMFD) on its treatment in children. **Methods** We performed a randomized clinical study comparing CMFD with cow's milk diet (CMD) in two groups each consisting of 70 patients (age range, 1-13 years) with chronic functional constipation (defined as Rome III criteria). All subjects had been referred to a pediatric gastroenterology clinic and had previously been treated with laxatives for at least 3 months without success; also all 140 patients performed skin prick test. The case group received CMFD for 4 weeks. After that they received CMD for 2 extra weeks. The control group received CMD for whole 6 weeks. A response was defined as decreased in signs and symptoms that not fulfilled Rome III criteria after 4 weeks of CMFD and came back to Rome III criteria after 2 weeks of CMD challenge. **Findings** After 4 weeks 56 (80%) patients of the case group responded in comparison to 33 (47.1%) patients in the control group ($P=0.0001$). In the case group after 2 weeks challenge 24 out of 56 (42.8%) responders developed constipation according to Rome III criteria. With other words, the frequency of cow's milk allergy among constipated patients was 80%. Only one patient had positive skin prick test. **Conclusion** In children, chronic constipation can be a manifestation of cow's milk allergy. At present, although several aspects must be further investigated, a therapeutic attempt with elimination diet is advisable in all children with constipation unresponsive to correct laxative treatment.